Authentik (IDP) integration with Nile

This document provides a comprehensive overview of integrating Authentik as an Identity Provider (IdP) with Nile. It outlines the necessary steps and technical configurations for a seamless integration experience.

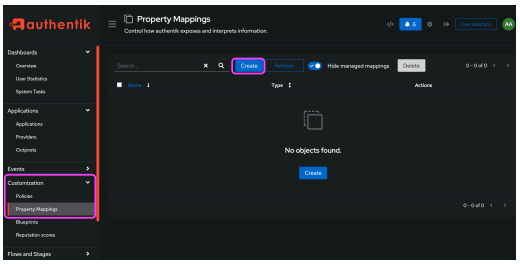
Complete the following configurations after setting up the Cloudflare tunnel and deploying the Authentik Docker container **Setup Cloudflare Tunnel with a Linux Server**

Requirements:

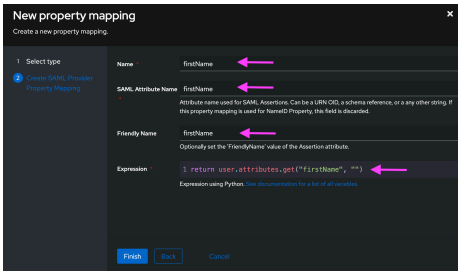
- Administrator rights to the Nile Portal.
- Administrator rights to Authentik

Step 1: Create Custom Property Mappings

1. Login to authentik.<domain> (For eg: authentik.nilenetworks.com)
2. Go to "Admin Interface" on the top right corner.
3. Navigate to Customization on the left panel and click "**Property Mappings**"



4. Select **SAML Provider Property Mapping** from the list
5. Create the **five** mappings listed below, each with its respective attributes.



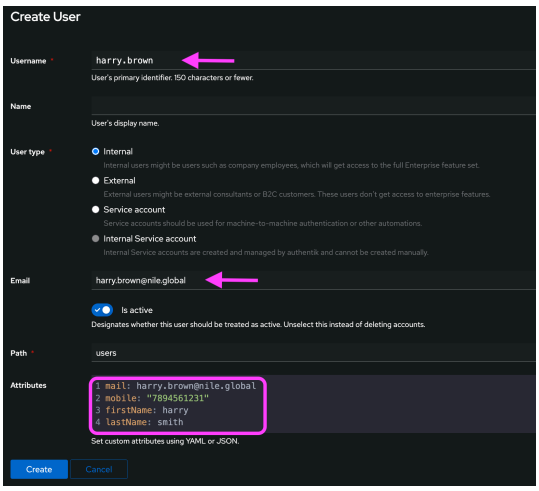
firstName	**return user.attributes.get("firstName", "")**
lastName	**return user.attributes.get("lastName", "")**
mail	**return user.attributes.get("mail", "")**
mobile	**return user.attributes.get("mobile", "")**
memberOf	**for group in request.user.ak_groups.all():** **yield group.name**

Step 2: Create User

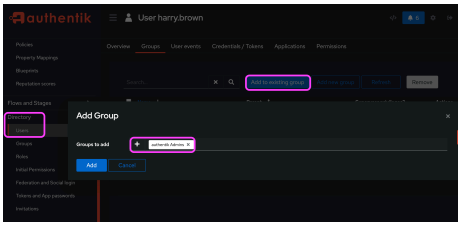
Now create admin users in Authentik.

1. Navigate to **Directory > Users**.
2. Click **Create** to add a new user.
3. Enter the required user details and attributes (mandatory). See the example below.

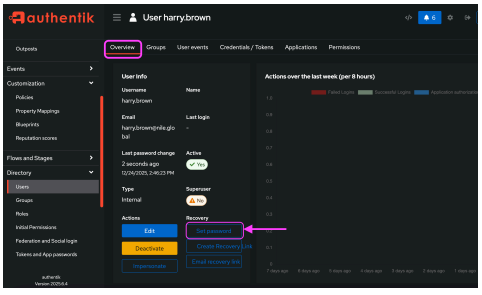
mail: harry.brown@nile.global
mobile: "7894561231"
lastName: smith
firstName: harry



4. After creating the user, map the user to the "**authentik Admins**" group.

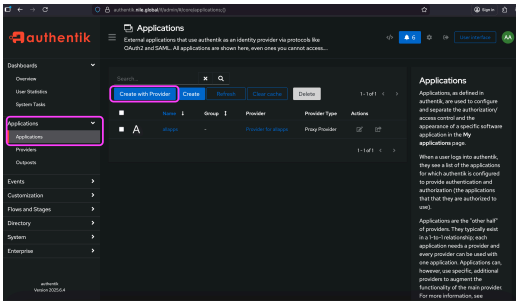


5. Set a new password for the user.



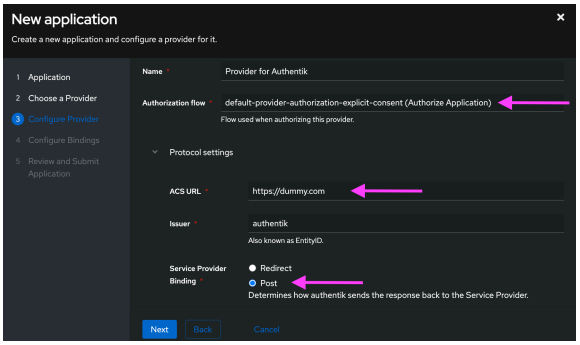
Step 3: Create Application

1. Navigate to "Applications" on the left side panel and again click "Applications" from the drop down.
2. Click on "Create with Provider" to create an application.

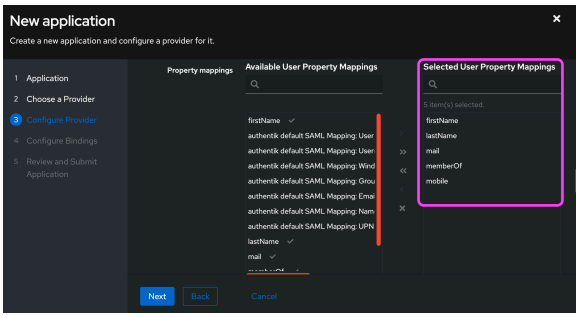


3. Provide a name for the application (e.g., "Authentik") and click **Next**.
4. On the next page, select the **SAML Provider** icon.
5. Configure the following parameters exactly as shown:

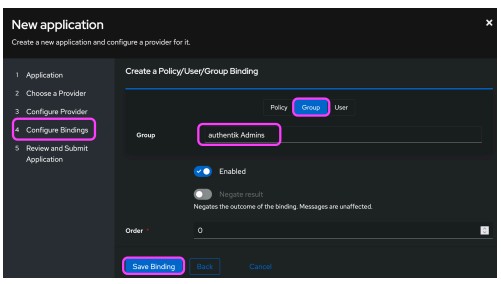
- **Authorization flow**: default-provider-authorization-explicit-consent (Authorize Application)
- **ACS URL**: https://dummy.com (temporary placeholder)
- **Service Provider Binding**: POST



6. Scroll down and expand **Advanced Protocol Settings**.
   Set **Signing Certificate** to **"authentik Self-signed Certificate"** from the drop-down menu.
7. Select the **Property Mappings** created in Step 1



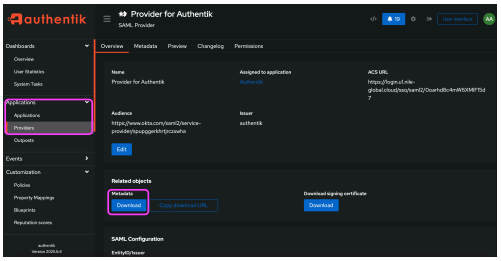
8. Choose "NameID Property Mapping" - **authentik default SAML Mapping: Email,** then click **Next**
9. At the Configure Bindings field - Choose the existing **Authentik Admins** group, then click Next and **Submit**



Step 4: Download the MetaData

Download the metadata file to configure in the Nile portal IdP settings

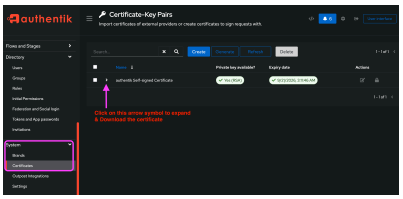
1. Navigate to **Providers** under **Application**
2. Click the "**Download**" button under "**MetaData**"



3. Open the downloaded xml file in a text editor.
4. Search (Ctrl+F) for sso/binding/post/ – this is your SSO URL.
   Copy the complete URL from Location=" to the closing " (including https:// and trailing /).
   Eg: **https://authentik.nilenetworks.com/application/saml/authentik/sso/binding/post/**
5. Search (Ctrl+F) for "entityid" and safeguard these parameters.

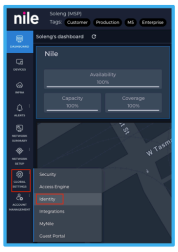
Step 5 : Download the self signed Certificate

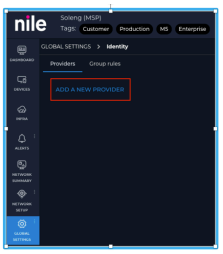
1. Navigate to System and then Certificates under the drop-down.
2. Click on the Arrow and download the self-signed certificate



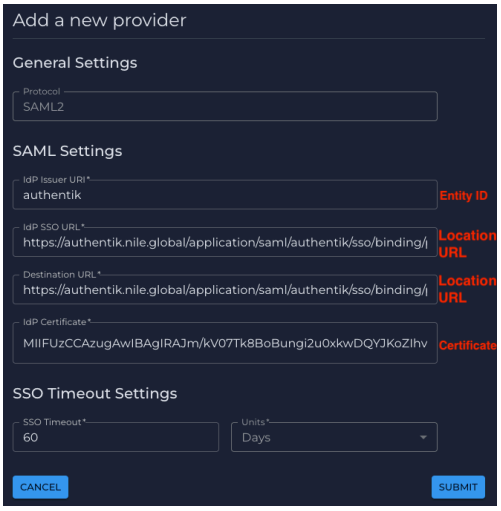
Step 6: Nile Portal Identity Provider Setup

1. Log in to the Nile Portal with the same Okta administrator at https://www.nile-global.cloud
2. Navigate to **GLOBAL SETTINGS** -> **Identity** to access the **Providers** page
3. Click on **ADD A NEW PROVIDER** to start configuring Okta as an IdP

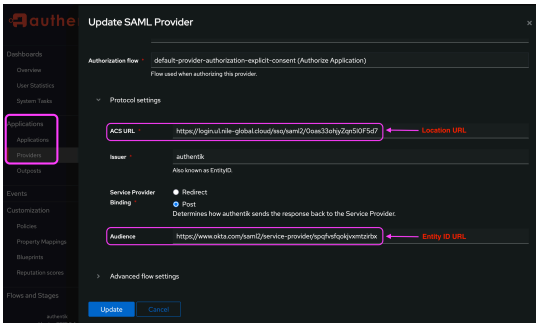




4. Fill out the 'Add a new provider' form from the data saved in the previous step:
   IdP Issuer URI: <Identity Provider Issuer>
   IdP SSO URL: <Identity Provider Single Sign-On URL>
   Destination URL: <Identity Provider Single Sign-On URL>
   IdP Certificate: Copy and paste the contents of the X.509 certificate previously downloaded



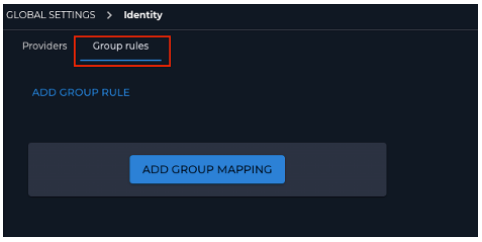
5. Click **SUBMIT** button to complete adding Authentik as an IdP provider.
6. Click on the **METADATA** button to download the xml metadata file
7. Open the downloaded metadata file with a text editor and collect the **entityID** and **Location** Their values will be used to replace the dummy parameters in the **Authentik app** and complete its setup.
8. Go to **Applications -> Providers** -> Edit the Provider for authentik and replace these values.



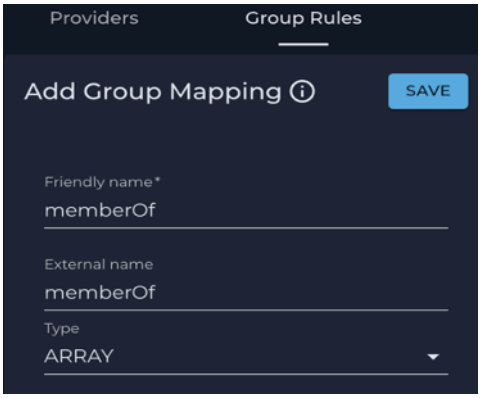
Step 7: Add Group Mappings in Nile portal

The following steps will configure Nile Portal Group rules to map Authentik groups to Nile Portal groups:

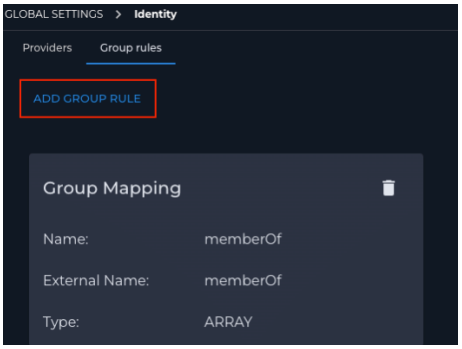
1. Click on the **Group Rules** tab, then click on the **ADD GROUP MAPPING** button to create a ‘memberOf’ Group Mapping rule.



2. Add the ‘memberOf' string in both the ‘Friendly name’ and ’External name’ fields, and select the ‘Type’ as ARRAY from the dropdown list, as illustrated:
3. Click the **SAVE** button to complete the GROUP MAPPING



4. Click on the **ADD GROUP RULE** link to start creating group rules

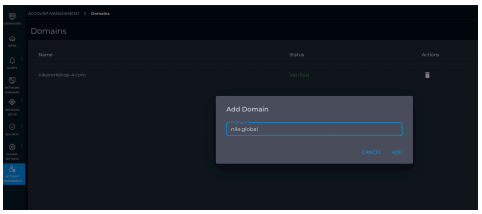


5. Create the rule for Nile Administrator and mention the Authentik group name under the value field.
6. Once the two rules are created, click the **INACTIVE** button for each rule to change it to **ACTIVE**:

Step 8: Add the domain in Nile Control Center.

Add the domain for the user created in Step 2. For eg. nile.global.

1. To add a new allowed domain, please navigate to **Account Management** >> **Domains** >> **Add Domain.**
2. Once the domain is added, click "**Verify**."



With the above steps, Authentik has been successfully integrated as the SSO provider for the Nile portal. To verify the setup, navigate to **https://u1.nile-global.cloud/** and sign in using the test user created at the beginning of this guide. When you attempt to log in, you will be redirected to the Authentik login page, and upon successful authentication you will be returned to the Nile portal with SSO enabled